Elements, Compounds & Mixtures

Exam code: 0620 & 0971

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Updated on 4 November 2025

Elements, compounds & mixtures

- All substances can be classified into one of these three types
  - Elements
  - Compounds
  - Mixtures

What is an element?

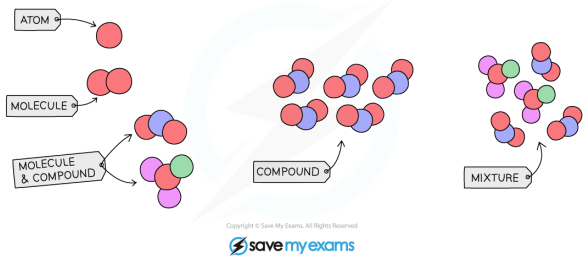
- A substance made of atoms that all contain the **same number of protons** and cannot be split into anything simpler **by chemical means**
- There are around 118 elements in the Periodic Table

What is a compound?

- A pure substance made up of two or more elements **chemically combined**
- There is an **unlimited** number of compounds
- Compounds cannot be separated into their elements by physical means
- E.g. copper(II) sulfate ($CuSO_4$), calcium carbonate ($CaCO_3$), carbon dioxide (CO_2)

What is a mixture?

- A combination of two or more substances (elements and/or compounds) that are **not** chemically combined
- Mixtures can be separated by **physical methods** such as filtration or evaporation
- E.g. sand and water, oil and water, sulfur powder and iron filings



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